



XXXVII SISFA National Congress - Bari 2017

Programme

26 September 2017 - Tuesday

15.00-18.35 - Palazzo Ateneo, Salone degli Affreschi - Bari, Piazza Umberto I

15.00 *Registration*

15.30 *Opening remarks by institutional representatives*

Invited talks

16.00 **Mauro Di Giandomenico**

Fifty years of History of Science in Bari

16.45 **Bruno Ghidini**

The historical roots of the Department of Physics of Bari
(provisional title)

17.30-17.50 Coffee break

17.50 **Salvatore Esposito**

Darwin and the others: the reception of Poli's Testacea outside Italy and other recent discoveries about the Molfetta scientist

18.35 *End of session*

18.45 Special evening event: Guided visit in old Bari city

27 September 2017 – Wednesday

9.00-13.25 – Dipartimento Interateneo di Fisica “Michelangelo Merlin”,
Campus Universitario “Ernesto Quagliariello”

THEME: Physics and Astronomy in the 20th century

Invited talks

9.00 Nadia Robotti

The Italian Physicists and the Great War

Communications

9.45 Stefano Bordoni

The historical roots of new debates on classical electrodynamics

10.05 Christian Bracco

Albert Einstein and the Marangoni family

10.25 Mario Calamia, Monica Gherardelli

Exact time: the first scientific application of remote communications

10.45 Fabrizio Bònoli, Valeria Picazzi

Is really the Guido Horn d'Arturo's tessellated telescope the forefather of the modern large multi-mirror telescopes?

11.05-11.25 Coffee break

THEME: Physics between the 17th and 19th centuries

Communications

11.25 Flavia Marcacci

All the planets are related to the Sun. Riccioli and his spiralized skies

11.45 Grazia Zini, Susanna Bertelli, Chiara Beatrice Vicentini, Paolo Lenisa

Pietro Torquato Tasso (1765-1842), the inventor: stories of watches and various instruments between science and everyday life

12.05 Rosanna Del Monte, Azzurra Auteri

The kaleidoscopes applied to figurative arts: the genius of Paolo Anania De Luca

12.25 Roberto Mantovani

A physics outsider: the case of the amateur scientist Gaetano Spandri (1796-1859)

12.45 Lucia De Frenza

Paolo Volpicelli: a scientist's career in papal Rome

13.05 Marco Taddia

On the supposed discovery of terrestrial helium by Luigi Palmieri
Titolo

13.25-14.40 Lunch break

14.40-18.30 - Dipartimento Interateneo di Fisica "Michelangelo Merlin",
Campus Universitario "Ernesto Quagliariello"

THEME: Physics and other disciplines

Communications

14.40 Fausto Casi

Silent Cinema between emigration and return

15.00 Benedetta Campanile

Roberto Mario Fano: an Italian among architects of the Information

15.20 Luca Guzzardi

Epistemic Constraints

15.40 Laura Franchini

The birth of the Nobel peace prize and the dispute about the use of
a scientific innovation

16.00 Giancarlo Albertini, Anna Sicolo

The perception of flowing time: metaphors and mental disorders

16.20-16.40 Coffee break

THEME: From antiquity to Galilei

Communications

16.40 G. Boscarino, P. Di Mauro, A. Pagano, E.V. Pagano

Elements of Geometrical Calculation in Archimede: The laws of
statics

17.00 Renato Migliorato

On the origins of a scientific paradigm

17.20 Vincenzo Cioci

New insights about Galileo's research on the fall of bodies. A
review

17.40 Annual SISFA General Assembly

20.30 Social Dinner

28 September 2017 - Thursday

9.00-13.30 - Dipartimento Interateneo di Fisica "Michelangelo Merlin",
Campus Universitario "Ernesto Quagliariello"

THEME: Physics in the 20th century

Invited talks

9.00 **Paolo Rossi**

Physics in Pisa in the first half of the XX century: a reappraisal

Communications

9.45 **Marcella Palese**

Noether Theorems and conservation laws in field theory

10.05 **Marco Di Mauro, Salvatore Esposito, Adele Naddeo**

A less known Feynman: an interplay between physics and biology at
the Hughes Aircraft Company

10.25 **Flavio Del Santo**

Karl Popper's engagement in quantum physics in the 1960s and his
critique of the Logic of Quantum Mechanics

10.45 **Antonino Drago**

Strocchi's formulation of Quantum Mechanics. Its improvement to an
alternative formulation to the dominant one

11.05-11.30 Coffee break

THEME: From the Middle Ages to the Modern era

Communications

11.30 **Maria Luisa Tuscano**

King Ruggero II and the Reform of the Calendar

11.50 **Ivana Gambaro**

Jesuit science and epistemology in 17th century

12.10 **Mauro Gargano**

Della Porta, Colonna and Fontana and the first astronomical
observations in Naples

THEME: Physics in the 19th century

12.30 **Tullio Aebischer**

Bicentenary birth A. Secchi 1818-2018: the geodetic activity

12.50 **Pasquale Tucci**

Angelo Secchi: "L'Unità delle Forze fisiche"

13.10 **Danilo Capecchi**

Christian epistemology of Gabrio Piola. The Lettere di Evasio a
Uranio

13.30-14.40 Lunch break

14.40-18.50 - Dipartimento Interateneo di Fisica "Michelangelo Merlin",
Campus Universitario "Ernesto Quagliariello"

Topical Session

Franco Selleri (1936-2013): His research and legacy on the foundations of quantum mechanics and relativity

Invited talks

14.40 Ludwik Kostro

□-Units and □-Mega Quantum of Action in their Historical Context

15.25 Gino Tarozzi

The reality of the wave function and of EPR's predictable properties in Franco Selleri's interpretation of Quantum Mechanics

16.10 Francesco Guerra

Franco Selleri: the courage of ideas

16.30-16.50 Coffee break

Communications

16.50 Giancarlo Scalera

Can the history of the ether receive new boost from geosciences?

17.10 Luigi Romano

Franco Selleri and His Contribution to the Research of Foundations of Relativity Theory

17.30 Angela Laurora

John Bell's scientific correspondence in the early 1970s

17.50 Angelo Baracca, Flavio Del Santo

Franco Selleri and Karl Popper: fighting side by side for realism in physics

18.10 Saverio Pascazio

Franco Selleri's Legacy and the Bell Inequality

18.30 Alexander Afriat

The circularity of rotation

18.50 End of sessions

29 September 2017 – Friday

9.00-13.20 - Palazzo Ateneo, Salone degli Affreschi - Bari, Piazza Umberto I

THEME: Didactics of Physics, Physics and society

Invited talks

9.00 Matteo Leone

Physicists in the Senate of the Kingdom of Italy:1848-1943

Communications

9.45 Gaetano Abate

The Science Museum of Irpinia in Avellino: recovery, restoration, record and didactics of physics

10.05 Daniele Buongiorno, Marisa Michelini

The conceptual contribution of the history in learning physics: the case of optical spectroscopy

10.25 Pietro Cerreta

Historical experiences and fundamental concepts of electrostatics

10.45 Lucio Fregonese

Coulomb's electrostatic law in Italian secondary-school textbooks. A difficult situation and possible solutions

11.05 Elena Corradini

The Italian University Museums Network for the promotion of the physics collections: development prospects

11.25-11.45 Coffee break

11.45 Panel discussion(coordinated by **Augusto Garuccio**):

The role of the history and didactics of physics and astronomy in the training of future physics teachers

-**Ileana Rabuffo** (Presidente Commissione Permanente Didattica - SIF)

-**Alberto Meroni** (Associazione per l'Insegnamento della Fisica - AIF)

-**Maria Mellone** (Unione Matematica Italiana - UMI)

-**Ginevra Trinchieri** (Presidente Società Astronomica Italiana - SAIIt)

-**Francesca Monti** (Rappresentante area fisica nel CUN)

13.20 Congress ending remarks

13.30-14.30 Lunch break

14.30 Special event: Cultural trip Trani (Cathedral, Synagogue and Hebrew Museum, Typewriters Museum)